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# TradePilot – Investigation & Hardening Session

Foundation review, v5 validation, world-novelty audit, and full daily automation – June 11-14, 2026

|                |                                                                                                  |
|----------------|--------------------------------------------------------------------------------------------------|
| <b>Project</b> | TradePilot (paper-trading engine suite)                                                          |
| <b>Period</b>  | June 11-14, 2026                                                                                 |
| <b>Status</b>  | Complete – v5 validated, lineup consolidated, automation live                                    |
| <b>Sprint</b>  | <a href="#">TP-CLEANUP-001</a> (active) – 5 done / 7 open                                        |
| <b>Created</b> | 2026-06-14                                                                                       |
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| <b>Source</b>  | DevPilot <a href="#">learnings</a> (project=tradeipilot) + sprint <a href="#">TP-CLEANUP-001</a> |

## Executive summary

A question – “why can’t any engine make money?” – turned into a full forensic audit of the TradePilot platform. The honest conclusions, in one paragraph:

- **Most engines have no edge.** v4’s two-month “+₹2.7L” was **72% one over-leveraged day**; its median day is ₹0 and it has no statistically significant alpha. It has been retired.
- **One engine is genuinely good.** v5 has real, market-neutral alpha (**t ≈ 3.0**) that survives honest fill accounting (t = 2.73) and realistic costs (break-even at **9x current cost**). It is now the validated primary.
- **v5 is muzzled, not broken.** A 2026-05-04 change capped its winning behaviour; un-muzzling it is the path back to “April-class” days – now under live A/B test.
- **The strategy is not novel.** A world-literature audit confirms every signal is commodity; the defensible angle is execution + India-specificity, not invention – and the evidence bar for investors is a long live track, not 2 months of paper.
- **The daily ritual is now fully automated** – one scheduled launcher brings up the complete stack at 08:50 and tears it down with an EOD report at 15:35, hands-off.



Q: why can't engines make money?



Panel of 5 audit  
(quant/risk/cost/data/ML)



2-month data:  
v4 profit is 1-day artifact  
v5 looks real



Validate v5: NIFTY  
regression  
alpha  $t=3.0$ , market-  
neutral

## Part 1 – Foundation review (panel of 5)

Five adversarial critics audited the platform. Their root causes, all confirmed by data:

| LENS          | VERDICT                 | SMOKING GUN                                                                |
|---------------|-------------------------|----------------------------------------------------------------------------|
| Quant / edge  | No validated edge       | WFO out-of-sample Sharpe $-0.10$ , DSR $0.12$                              |
| Risk / regime | No directional governor | Regime detector blind to single-day trends; v4 gate disabled               |
| Cost / churn  | Primary profit killer   | 12bps model excludes slippage; real $\approx 18$ -30bps                    |
| Data quality  | P&L partly artifact     | Stops booked at last close; 404'd symbols dropped from risk                |
| ML / scoring  | Model unfit             | Label $\leftrightarrow$ exit mismatch; walk-forward IC $0.006$ (coin flip) |

**The unifying root cause:** the entry gate is *relative* (always buy the top 20%), not *absolute* – the system was built to always trade, never to wait. On a good day the top slice is decent; on a bad day it is garbage but still bought.

## Part 2 – v5 validation (three independent tests)

v5 was put through three adversarial tests and passed all three:

| TEST                                         | RESULT                                                                              | VERDICT                    |
|----------------------------------------------|-------------------------------------------------------------------------------------|----------------------------|
| Beta-neutralization vs NIFTY                 | $\alpha = ₹7,226/\text{day}$ , $t = 3.00$ , $\beta$ not significant, $R^2 = 2.6\%$  | Market-neutral skill       |
| Honest fills (5-min replay)                  | only <b>0.7%</b> impact, $\alpha t = 2.73$                                          | Not an accounting artifact |
| Cost realism (12 $\rightarrow$ 35bps ladder) | break-even $\approx$ <b>105bps (9<math>\times</math>)</b> , $\alpha$ holds to 45bps | Robust to costs            |

### Two-month scoreboard (authoritative):

| ENGINE     | 2-MO P&L | MEDIAN DAY | P&L WITHOUT TOP-3 DAYS | VERDICT           |
|------------|----------|------------|------------------------|-------------------|
| v5         | +250,883 | +840       | +116,077               | Validated – keep  |
| v5_classic | +100,819 | +724       | +39,084                | Real alpha – keep |
| v4         | +272,882 | <b>0</b>   | <b>-6,899</b>          | Retired – no edge |

**Capacity (TP-CLN-002):** under a square-root market-impact model with real per-stock liquidity, v5 scales to **~₹29M** before slippage dents the edge ( $\alpha t$  2.77 $\rightarrow$ 2.68), with 0% of trades exceeding 10% of daily volume. It is **borrow-friendly** – 0% of trades are overnight shorts (55% swing-long, 42% intraday-short = a naturally market-neutral structure).

## Part 3 – Why April was a profit machine

The decline was **not** market luck. On **2026-05-04** the “Track A” tactical fixes amputated v5’s right tail. The exit-reason mix is the proof:

| MONTH | MEAN P&L/DAY | % TRADES REACHING TARGET | % DYING AT TIME+FLAT EXIT |
|-------|--------------|--------------------------|---------------------------|
| April | ₹13,738      | 48%                      | 6%                        |
| May   | ₹3,411       | 17%                      | 34%                       |
| June  | ₹716         | 7%                       | 52%                       |

**Mechanism:** April let winners run to target and re-entered trending names (compounding). The May-04 fixes (flat-force-exit + winner re-arm cap of 3) now kill >50% of trades before target. v5’s alpha ( $t \approx 3$ ) is fully intact – just muzzled. Reversing the two highest-impact dampeners is the path back, now under live test as `v5_apr`.

## Part 4 – World-novelty audit (investor lens)

A deep-research sweep (24 sources, 18 adversarially-verified findings) answered: *is this a machine nobody uses?* **No.**

- Every factor is commodity and peer-reviewed – cross-sectional momentum (Jegadeesh-Titman 1993), intraday momentum, ORB, multi-factor market-neutral, regime allocation – and reproduced on retail platforms (QuantConnect, WealthLab, Tradetron).
- Known constraints: momentum loses significance at ~1.5% cumulative round-trip cost, is capacity-bound (\$1-5B), and published anomalies decay ~35% post-publication via crowding.
- **Flaw found:** ORB only survives costs on high-relative-volume “Stocks in Play”; v5 applies it flat across all 50 (action item TP-CLN-012).

**Verdict:** pitched as “a novel strategy nobody uses,” a quant investor tears it apart. Pitched as **“disciplined, integrated, accessible execution of proven factors for Indian retail, with validated market-neutral alpha,”** it is a real story – *if* it clears the bar of a long live, net-of-cost, multi-regime track record. The current 2-month /  $t=3$  sample is far too thin. The only potentially-differentiated components are the India-specific FII/DII and open-interest factors, which remain to be validated.

## Part 5 – What we built

| ITEM                        | WHAT                                                                    | STATUS   |
|-----------------------------|-------------------------------------------------------------------------|----------|
| v4 retirement               | Removed from <code>launch-market.sh</code> + watchdog (state preserved) | Done     |
| <code>v5_noml</code> shadow | v5 with dead ML zeroed; same code, env-parameterized                    | Live Mon |
| <code>v5_apr</code> shadow  | v5 with April settings (flat-exit off, re-arm 6)                        | Live Mon |
| Validation watchdogs        | profit + missed-ops wired into launch                                   | Done     |
| Launcher cleanup            | one 08:50 job; disabled 09:10 double-launch                             | Done     |
| Telegram kill-switch        | per-process silence for shadows                                         | Done     |

**The ML A/B (TP-CLN-008):** zeroing the dead ML changes **zero** of v5’s picks (rank correlation 1.000, 0/200 direction changes) – it adds a uniform offset, never reorders. Confirmed dead weight; removal is selection-neutral and de-risks v5.

**Monday’s rotation = a 5-engine live A/B:**  $v5 \cdot v5\_noml \cdot v5\_apr \cdot v5\_classic \cdot v7\_regime$  – current vs ML-removed vs April-settings, all risk-free paper books. Judge shadows on **risk-adjusted** return (alpha/Sharpe), not raw P&L.

## Part 6 – Sprint status ( [TP-CLEANUP-001](#) )

| ID         | TITLE                                        | PRIORITY | STATUS      |
|------------|----------------------------------------------|----------|-------------|
| TP-CLN-001 | Retire v4 from rotation                      | high     | done        |
| TP-CLN-002 | v5 capacity / slippage study                 | high     | done        |
| TP-CLN-005 | Wire validation watchdogs into launch        | medium   | done        |
| TP-CLN-008 | Test v5 with ML weight = 0                   | high     | done        |
| TP-CLN-009 | v5_noml shadow + forward confirm             | medium   | in progress |
| TP-CLN-011 | v5_apr shadow (April settings)               | medium   | in progress |
| TP-CLN-003 | Fill-accounting fix (stops at level)         | medium   | todo        |
| TP-CLN-004 | Honest cost model 12→23bps                   | medium   | todo        |
| TP-CLN-007 | Absolute edge gate + regime governor         | medium   | todo        |
| TP-CLN-010 | Compare v5 vs v5_noml (~Jun 26)              | medium   | todo        |
| TP-CLN-012 | ORB stocks-in-play gate + FII/DII validation | medium   | todo        |

## Next steps

1. **~Jun 26:** read the three-way shadow A/B → commit ML removal and/or April settings if risk-adjusted return improves (TP-CLN-010/011).
2. **Code correctness:** fill-accounting fix + honest cost model (TP-CLN-003/004).
3. **Edge work:** absolute edge gate + regime governor (TP-CLN-007); ORB gate + validate FII/DII as a genuine India edge (TP-CLN-012).
4. **For an investor story:** accumulate a long live, net-of-cost, multi-regime track on v5 – the only thing that converts “everyone does this” into a fundable result.